

PhotoEcho: Facilitating Intergenerational Dialogue Through AI-Enhanced Family Photograph Narratives

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Abstract

Intergenerational communication is crucial in promoting family emotional bonding and cultural inheritance, but the accelerated pace of life and differences in values in modern society have gradually alienated grandchildren and grandparents from each other. Photo narratives are an effective way of communication, which can visually and vividly convey information and emotions, and are easy to be understood and resonated by the audience. In this paper, we propose PhotoEcho, an intelligent application based on old photo narratives, to promote the emotional communication and connection between grandchildren and grandparents. Based on image recognition and generative AI technologies, the tool can assist photo storytelling, support dynamic content generation, and guide good communication between grandparents and grandchildren and family culture transmission. Our user study shows that PhotoEcho not only significantly improves the quality and depth of intergenerational communication, but also promotes emotional connection and memory sharing, providing an innovative path for family cultural inheritance in the digital age.

CCS Concepts

• Human-centered computing → Interaction paradigms.

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Keywords

Intergenerational Communication; Photo Narratives; Generative AI; Emotional Connection; Family Cultural Heritage

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1 Introduction

Grandparent-grandchild relationships play a crucial role in family emotional connections, cultural transmission, and identity formation [3] [16] [5], but modern fast-paced lifestyles, geographical distances, and the digital divide have increasingly limited intergenerational communication [2] [15]. Photographs, as visual memory carriers spanning time and space, have the potential to evoke emotional resonance and facilitate communication [7]. Human-computer interaction research indicates that technology can enhance family memory narratives. Bumsoo Kang et al. [9] explored the memory-preserving role of interactive digital photo albums; Ziyu Wan [21] investigated how AI image enhancement improves the readability of old photos. However, these studies primarily focus on image restoration or one-way narratives, lacking support for two-way emotional dialogue. This study proposes PhotoEcho—a generational communication system integrating generative AI—centered on three mechanisms: AI photo analysis and story prompts, grandchild questioning support to break one-way narratives, and transforming photos and oral histories into immersive videos. Mixed-method assessments show that PhotoEcho significantly enhances the depth of intergenerational dialogue, emotional

connection, and satisfaction with cultural inheritance. The contribution of this study is:

1. Design and implement an AI-integrated intergenerational communication system.
2. Propose a “two-way photo narrative” framework to expand family memory-sharing theory.
3. Provide empirical evidence that AI-enhanced narratives improve intergenerational communication quality.

2 Related Work

2.1 Photo-based intergenerational narratives

Intergenerational narratives can alleviate the intergenerational gap, promote family understanding and support, and play an important role in cultural transmission [6] and value transfer [17] [4]. Photographs are crucial in intergenerational narratives as visual time markers and memory media that evoke past events and emotions. Keightley and Pickering [11] pointed out that photo narratives strengthen family emotional ties and help younger generations understand family history and cultural context. Grandparents convey family traditions through the display of old photographs and oral narratives, and grandchildren participate with questions, responses, and associations, creating a two-way interactive narrative. For example, Story-Me [13] enhances narrative collaboration between different age groups through differentiated device and interface design.

2.2 Digital technology empowers photo-based intergenerational narratives

To enhance photo-based narrative experiences, researchers have explored a variety of digitally mediated technologies, including augmented reality and virtual reality. For example, StoryBox [20] combines a physical interactive device with a digital screen to encourage grandchildren to browse and tell family stories together, which enhances immersion in face-to-face communication. Grandtotem [14] uses AR to embed old photos into a virtual scene, encouraging family members to explore the history together. Wei et al. [22] explored how VR can be used for remote grandchildren interactions to alleviate spatial disconnection.

Although existing systems are innovative in media, they are mostly limited to fixed scenes or contents, making it difficult to dynamically perceive user emotions and contexts. For this reason, AI technology is introduced to enhance the flexibility and immersion of narrative guidance and interaction.

2.3 AI-driven intergenerational communication

Advances in natural language processing, image recognition, and other fields have increased the potential of AI in intergenerational communication, and AI can optimize the interaction experience and strengthen the emotional connection through dialogue guidance and sentiment analysis [19]. For example, MAMAS [8] optimizes family communication by evaluating parent-child dining conversations in real time and providing feedback through voice sentiment analysis, Zaturi [10] helps working parents efficiently create

audiobooks by capturing “micro-time” based on sensors, balancing emotional transfer and time management, and Dyadic Mirror [12] enhances parent-child self-reflection and emotional resonance by using visual recognition and second-person feedback.

Unlike the above systems that focus on behavioral perception and feedback, PhotoEcho fuses image recognition, voice interaction and LLM technology to support grandparents and grandchildren to collaboratively generate video narratives and inspire personalized intergenerational conversations.

3 Design Goals

In order to investigate the current state of communication between grandparents and grandchildren as well as their needs and requirements, the study conducted a questionnaire survey with 10 children aged 9–12 (Mage=10.50, SDage=0.87) and semi-structured interviews with 10 grandparents (Mage=69.00, SDage=3.46). The content covered communication frequency, mode, content, barriers and needs. All processes followed the principles of informed consent and privacy protection. The results revealed five major barriers to intergenerational communication: low frequency, single mode, lack of topics, differences in language understanding and cultural divides, which constrain the depth of interaction. Accordingly, three design inspirations are proposed:

1. To develop a low-threshold intergenerational communication and emotional connection tool.
2. To build a platform for sharing common topics and memories across generations based on old photos.
3. To utilize AI-assisted technology to promote interactive dialog experiences between grandchildren and grandparents.

4 PhotoEcho Design

4.1 Function and Interaction Design

PhotoEcho runs on the tablet side. The core functionality consists of grandparents and grandchildren starting a dialog based on old photos and working together to create a video. The system also includes an AI voice assistant to guide the communication, and the usage flow is shown in Figure 1.

Users can take or select old photos from photo albums to narrate (Figure 2a). The system automatically recognizes the photo elements and generates interaction icons (Figure 2b, 2c). Clicking on the interaction icon pops up a communication prompt panel to assist grandchildren in communicating (Figure 2d), and the AI voice assistant guides grandchildren to narrate the photo story (Figure 2c). During the process the grandchildren can click on the photo annotations to ask questions or participate in a discussion with the grandparents (Figure 2d). After the conversation, the grandchildren can upload their personal photos to the system (Figure 2e). The system generates a video based on the photos and storytelling to preserve the grandchildren’s exclusive memories (Figure 2f).

4.2 AI-assisted intergenerational narrative and video generation

4.2.1 Photo Element Recognition and Segmentation Labeling.

The user selects input images by camera shots or photo albums. The images are fed into a pre-trained Mask R-CNN model for element

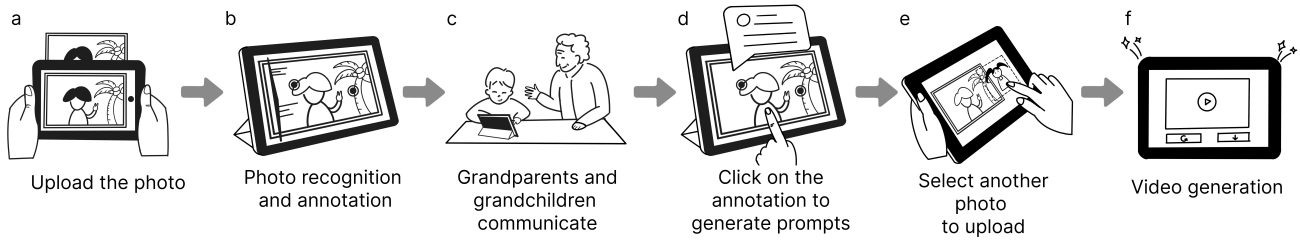


Figure 1: Use flow.

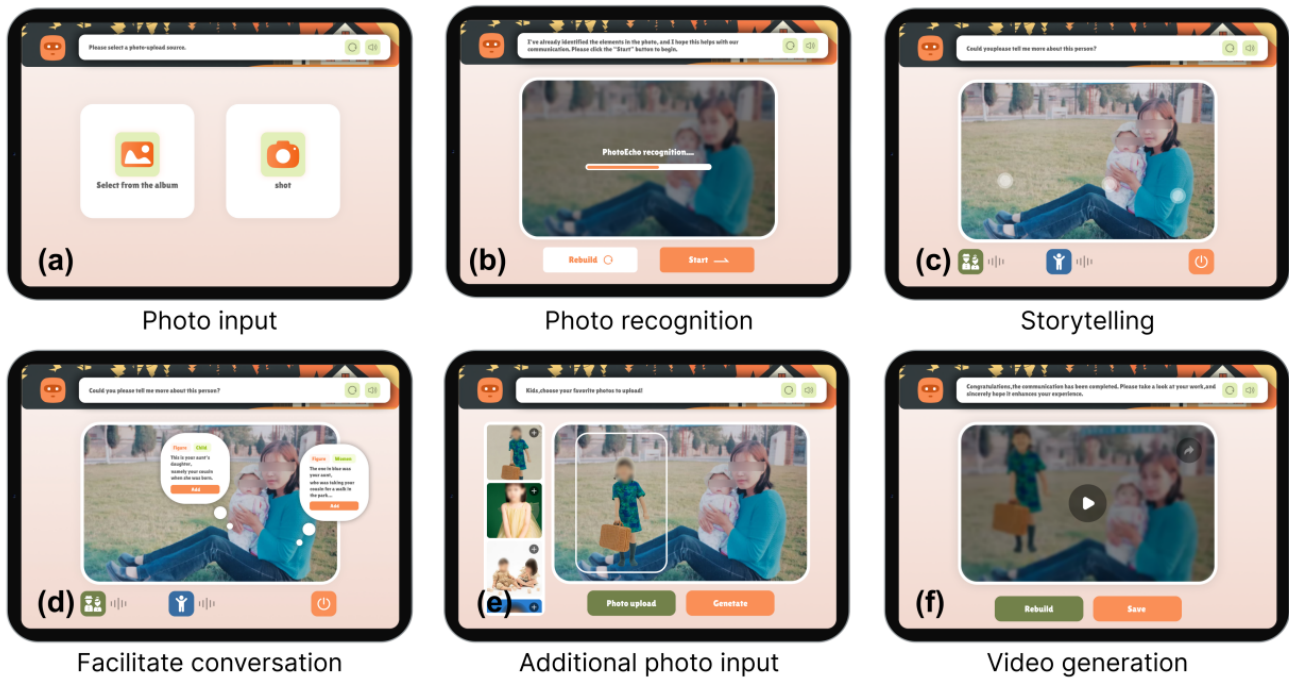


Figure 2: PhotoEcho application design.

recognition and labeling generation. The results are presented in a visualized form, including an interaction icon (Figure 2c), and a panel of object properties displayed by clicking the interaction icon (Figure 2d).

4.2.2 Dynamic Video Generation.

The speech recognition component converts the user-input speech into textual information and inputs it into the dialog management system to realize real-time updating of contextual information. Then, with the help of a large language model, it realizes the automatic generation of narrative scripts. Finally, the Gen-3 Alpha model is invoked to generate short videos (3-5 seconds) with basic actions.

5 User Study

5.1 Participants and Process

Participants. To evaluate the effectiveness and user experience of PhotoEcho in facilitating intergenerational communication between grandparents and grandchildren, we recruited 12 pairs of participants from real grandparent-grandchild relationships, with the average age of grandparents being 66.92 years old ($SD_{age}=3.94$) and the average age of grandchildren being 11.17 years old ($SD_{age}=1.53$). All participants came from real families and were provided with an old photograph of family significance, and grandparents were required to have the ability to narrate the story of that photograph. All participants signed an informed consent and privacy protection agreement. Participants were randomly divided into two groups: the experimental group (6 pairs) used the PhotoEcho application and the control group (6 pairs) did not use PhotoEcho and were provided with uniform written task guidelines (Table 1).

Table 1: Experimental and control group information

Group	N	Grandchildren		Grandparents		Distinction between groups
		Gender	Age	Gender	Age	
PhotoEcho	6 pairs	3M/3F	11.00	3M/3F	67.00	Using PhotoEcho
Baseline	6 pairs	3M/3F	11.30	3M/3F	66.83	Not using PhotoEcho

Procedure. The experiment was conducted in the participants' homes in a quiet environment over a 60-minute period and was divided into four phases: a) Preparation phase (10 minutes). Participants were explained the procedure, the experimental group was trained on the core functions of PhotoEcho, and the equipment was pre-commissioned. b) Pre-test phase (15 minutes). Participants completed a questionnaire based on the Inclusion of Other in the Self (IOS) scale [1] to assess the level of closeness between grandchildren and grandparents; c) Task execution phase (20 minutes). The experimental group used the PhotoEcho application to guide the photo discussion and generate a dynamic video, while the control group used a written guide to communicate freely, and the whole process was audio- and video-recorded; d) Post-testing phase (15 minutes). Semi-structured interviews with grandparents and grandchildren were conducted separately to collect feedback on task experience, affective changes, and functioning.

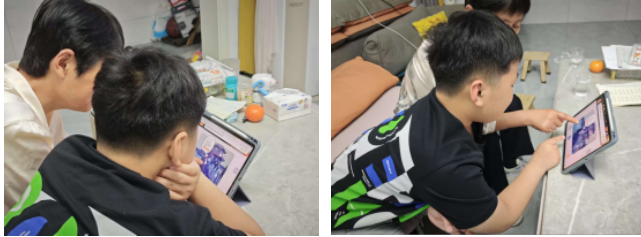


Figure 3: User study process where each column corresponds to one part of the study

5.2 Results

5.2.1 Improvement of Communication Quality and Efficiency.

We recorded the duration of conversations and the frequency of questions asked. We used the Sentence-BERT model to analyze semantic similarity [18] and used an independent sample t-test to assess conversational fluency and interaction frequency. The experimental group outperformed the control group in terms of conversation duration (17.2 vs. 11.5 minutes, $p < 0.05$), semantic relevance (0.78 vs. 0.59, $p < 0.05$), and question frequency (8.4 vs. 4.1 times, $p < 0.05$) compared to the control group, indicating higher content coherence. Grandchildren were able to ask follow-up questions around the elders' narratives under the system's guidance. AI annotations provided structured question assistance, while AI voice assistants enhanced the naturalness of the elders' expressions. Content analysis shows that PhotoEcho expands the dimensions of conversation. For example, after seeing a photo of her grandmother wearing a cheongsam, a grandchild used prompts to ask questions such as "Did everyone dress like that back then?" and "Did you take

videos back then?" demonstrating image-driven topic expansion; in contrast, the control group primarily asked closed-ended questions like "Where was this taken?" and "How old were you when this was taken?" lacking contextual extension.

5.2.2 Deepening of Emotional Connection and Empathy.

To assess the effect of PhotoEcho's emotional connection, we analyzed the interviews and task transcripts and found that: 83% of the participants in the experimental group felt that the other person was more focused and engaged, with one grandparent commenting, "Normally, I seldom take the initiative to talk about this, but with the prompting of the app, it felt more natural to me." ; 79% reported feeling a sense of understanding and respect for each other, with one grandparent mentioning, "I'm happy that my grandchildren talk to me about a lot of things they wouldn't normally talk to me about." The feedback indicates that PhotoEcho not only enables grandchildren to gain a deeper understanding of their grandparents' experiences, but also nurtures their empathy. One grandchild shared, "I used to think that my grandma's childhood was far away from me, but now I think she was cool when she was young." Another participant volunteered, "Next time, I would like to ask my grandpa to tell me about the pictures he used to take at work."

5.2.3 AI Ethical Risks and Privacy Concerns.

The semi-structured discussion revealed three ethical challenges of AI-assisted intergenerational communication. First, in terms of data privacy, 42.9% of the grandparents were concerned about the risk of cloud storage, such as "Will the voice be saved permanently?" Second, in terms of memory authenticity, grandparents' attitudes toward photo processing are divided, such as "seeing the child appearing in the scene as if we have been there together" versus "the generated dynamic scene is not too real"; and then, in terms of the risk of technological dependence, grandparents are worried that their grandchildren will rely on the prompts too much and lack active thinking. These findings highlight the need to balance technological innovation with ethical risks.

6 Limitations and future work

Although PhotoEcho performed well in enhancing grandchildren's emotional communication, there are still limitations. For example, the small sample size and lack of cultural diversity limit the generalizability of the results. In addition, AI-generated content may suffer from misidentification and information bias, and ethical safeguards need to be strengthened, such as incorporating a user confirmation mechanism, restricting the generation of sensitive information, and optimizing the logic of gender and cultural label recognition. In the future, functions such as two-way story co-creation and dynamic video input will be further explored to enhance the emotional depth and inclusiveness of interaction.

7 Conclusion

This study proposes the PhotoEcho system, which facilitates communication and emotional connection between grandparents and grandchildren through old photo narratives and AI generation technology. The system helps to build an interactive experience with depth through generative AI-assisted story guiding and labeling. Although there are still technical and ethical challenges, PhotoEcho demonstrates the potential of AI application in humanistic care scenarios. In the future, we will optimize the usage experience and expand the multicultural adaptability and functional extensibility.

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